

CPE 334 INTRODUCTION TO SOFTWARE ENGINEERING IN THE AGE OF AI AGENTS

Course: CPE 334 (Semester 1/2026)

Lab Increment: Lab 3 (60 Points)

Student Name: Natthakamol Katippatee

Student ID: 67070505215

Repository: natthakamol1130/toktickit

Peer Reviewer: Suprawi Sriamwong
(@Suprawi5227)

Peer Repository: Suprawi5227/toktickit

Submission Date: October 4, 2026

Answer Part 1: Git Use with Engineering Workflow

\$2

The TokTickIT Lab 3 engineering increment extends the single-context requester prototype into a production-grade enterprise IT Service Desk with multi-role authentication (JWT, bcrypt, mandatory password reset), role-based access control, IT Staff Ticket Queue and Internal Notes, and an Administrator User Management Console with safety rules.

\$2

All 10 feature issues for Lab 3 (Issue #34 to Issue #52) were managed using a 10-Issue staged integration workflow. Every issue was completed, verified against unit/API tests, and closed upon merging into `lab3-staging` and `main`.

29
30
31
32

1.6 Peer Review Evidence (5 คะแนน)

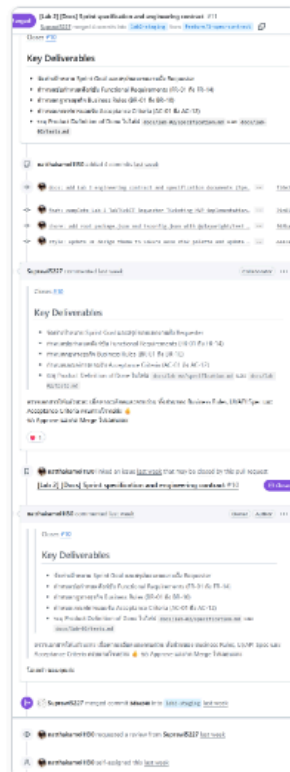
Author / Developer: Natthakamol Mornpam (นางสาวณัฏฐกมล มอญปาน — 67070505215) — GitHub: @natthakamol1130

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227) — GitHub: @Suprawi5227

Pull Requests I authored (reviewed by my partner @Suprawi5227) — รายละเอียดข้อความคอมเมนต์ที่รวบรวมเพิ่มเติม ห้ามย่อ:

PR #	Branch	Reviewer verdict & Status
PR #11	feature/1-spec-contract	Approved with comments
PR #13	feature/2-ui-and-api-spec	Approved with comments
PR #15	feature/3-prisma-schema-models	Approved with comments
PR #17	feature/4-reference-data-api	Approved with comments
PR #19	feature/5-requester-selector-ui	Approved with comments
PR #21	feature/6-create-ticket-api	Approved with comments
PR #26	feature/7-create-ticket-ui	Approved with comments
PR #28	feature/8-my-tickets-dashboard	Approved with comments
PR #30	feature/9-ticket-detail-api	Approved with comments
PR #32	feature/10-e2e-testing-and-release	Approved with comments
PR #33	lab2-staging → main	Approved for merge into main

PR #11: feature/1-spec-contract → lab2-staging



PR #13: feature/2-ui-and-api-spec → lab2-staging

```
# Logs & OS
*.log
.DS_Store
Thumbs.db
```

.gitignore

```
1  # dependencies
2  node_modules/
3
4  # env & secrets
5  .env
6  *.env
7  .env.local
8  .env.*.local
9  !.env.example
10
11 # build output
12 dist/
13 build/
14 client/src/**/*.js
15 client/tests/**/*.js
16
17 # uploads
18 server/uploads/*
19 !server/uploads/.gitkeep
20
21 # prisma
22 server/prisma/*.db
23
24 # IDE & Editor
25 .vscode/
26 .idea/
27 *.swp
28 *.swo
29
30 # logs & OS
31 *.log
32 .DS_Store
33 Thumbs.db
34
```

1.6 Peer Review Evidence (5 คะแนน)

Author / Developer: Natthakamol Mornpam (นางสาวณัฏฐกมล มอญปาน — 67070505215) — GitHub: [@natthakamol1130](#)

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสริณวัฒน์ — 67070505227) — GitHub: [@Suprawi5227](#)

Self-Review: I authored (created) my part of the project. I received feedback from my peer reviewer and I improved my work.

- Backend Vitest Integration Test Suites (`server/tests/lab-02/*.test.ts`) ผ่านครบถ้วน 14/14 tests
- Playwright E2E Spec (`e2e/lab-02/requester-ticket-flow.spec.ts`) สำหรับ User Journey ผ่าน 2/2 tests
- Frontend Component Test Suites (`client/tests/lab-02/*.test.tsx`) ครบทุกฟังก์ชันค่ะ

ฝากตรวจสอบอีกรอบได้เลยนะคะ ถ้ายอดเยี่ยมแล้วรบกวนกด Approve & Merge ได้เลยน้า ขอคุณมากค่ะ

Suprawi5227 commented last week

Collaborator



ตรวจสอบเรียบร้อยแล้ว ขออนุญาต Approve & Merge นะคะ สมบูรณ์แบบค่ะ



Suprawi5227 merged commit `a879a9e` into `lab2-staging` last week

PR #33: lab2-staging → main (Release PR)

Merged

[Lab 2] Release v1.0.0: Full Feature Implementation & E2E Validation #33

Suprawi5227 merged 51 commits into `main` from `lab2-staging`

Objectives & Scope

- ปลอยซอฟต์แวร์ Lab 2 เวอร์ชันสมบูรณ์ (Release v1.0.0) โดยหลอมรวมฟังก์ชันทั้งหมดจากสาขา lab2-staging เข้าสู่สาขาลึก main
- ผ่านการตรวจสอบสัญญาข้อตกลง สเปก UI Zen Green, REST API Endpoints และชุดการทดสอบการใช้งานแบบครบวงจร

Feature Summary

- Feature 1: กำหนดกรอบเป้าหมาย สัญญาข้อตกลงทางวิศวกรรม และ Definition of Done
- Feature 2: กำหนดมาตรฐาน Zen Green UI Design Tokens (#006B3C) และสัญญา REST API Endpoints
- Feature 3: ออกแบบ Prisma Models (RequesterUser, Ticket, Attachment) พร้อม Seed Data
- Feature 4: REST API สำหรับข้อมูลอ้างอิง (Requesters, Categories, Related Systems)
- Feature 5: หน้าจอพัฒนาสำหรับเลือกสิทธิ์ผู้ใช้งาน (Requester Selector Screen & Header Context)

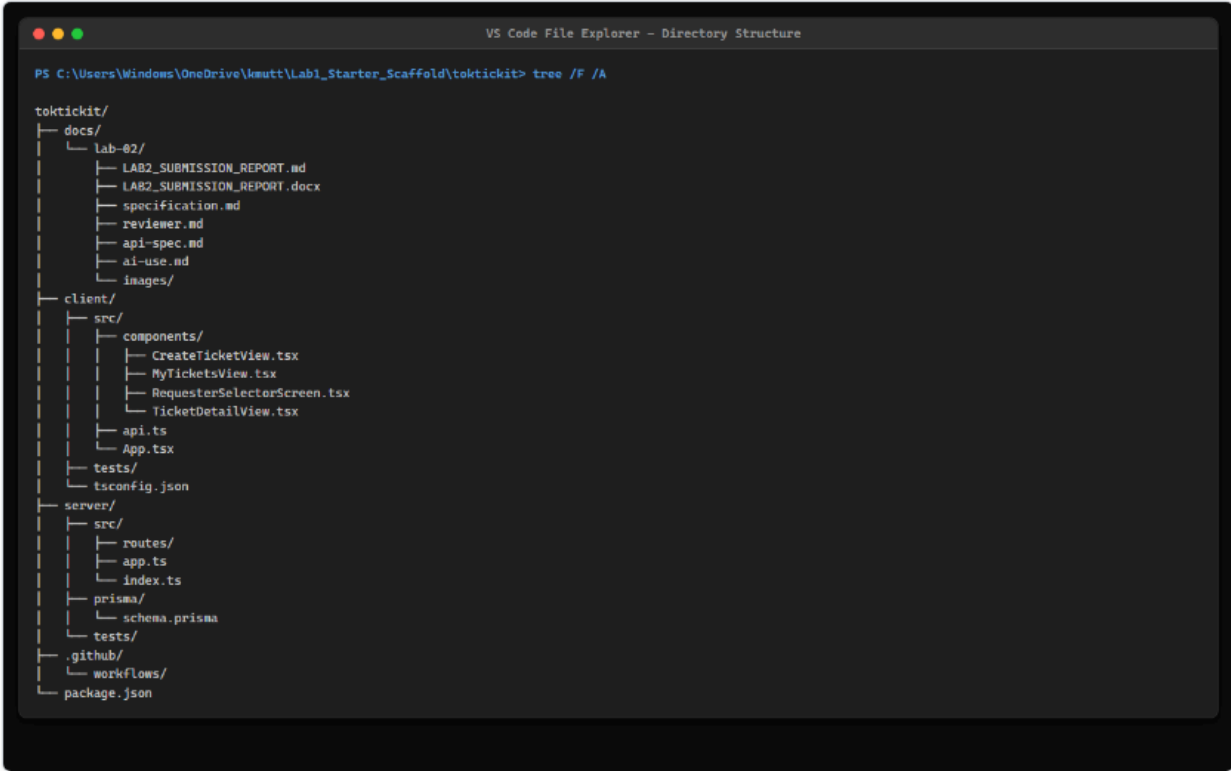
GitHub Project Kanban Board Part 2

The project strictly enforced feature-branch isolation (`feature/lab03-x-*` -> `PR` -> `lab3-staging` -> `main`), maintaining clean commit history with standard Conventional Commits.



- **Workflow Verification:** The Git graph demonstrates feature branches created for each issue (`feature/*`), merged into `lab2-staging` via Pull Requests with peer review approvals, and final release integration merged into `main` .

1.4 Repository Directory Structure



- **Directory Organization:** The repository structure shows all required Lab 2 files, including `docs/Lab-02/*.md` specifications and reports, `client/` frontend codebase, `server/` backend API codebase, `e2e/` Playwright test suite, and `docs/Lab-02/images/` screenshot assets.

Git Commit Log Graph

Table 1: Pull Requests Given (natthakamol1130/toktickit)

Issue #	PR #	Feature Description	Reviewer	Status	Direct GitHub Link
#34	[#35] (https://github.com/natthakamol1130/toktickit/pull/35)	Lab 3 Specification, UI Spec, and API Spec Documents	@Suprawi5227	Merged	[PR #35] (https://github.com/natthakamol1130/toktickit/pull/35)
#36	[#37] (https://github.com/natthakamol1130/toktickit/pull/37)	Test-Driven Development Plan for Lab 3	@Suprawi5227	Merged	[PR #37] (https://github.com/natthakamol1130/toktickit/pull/37)
#38	[#39] (https://github.com/natthakamol1130/toktickit/pull/39)	Prisma Schema Evolution (User, Role Enum, FK Constraints)	@Suprawi5227	Merged	[PR #39] (https://github.com/natthakamol1130/toktickit/pull/39)
#40	[#41] (https://github.com/natthakamol1130/toktickit/pull/41)	Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	@Suprawi5227	Merged	[PR #41] (https://github.com/natthakamol1130/toktickit/pull/41)
#42	[#43] (https://github.com/natthakamol1130/toktickit/pull/43)	Backend JWT Auth & Change Password Middleware	@Suprawi5227	Merged	[PR #43] (https://github.com/natthakamol1130/toktickit/pull/43)
#44	[#45] (https://github.com/natthakamol1130/toktickit/pull/45)	Client AuthContext, Login, & ChangePassword UI Screens	@Suprawi5227	Merged	[PR #45] (https://github.com/natthakamol1130/toktickit/pull/45)

#46	[#47] (https://github.com/natthakamol1130/toktickit/pull/47)	Requester Ticket & Comment Access Control APIs	@Suprawi5227	Merged	[PR #4 (https://github.com/natthakamol1130/toktickit/pull/47)
#48	[#49] (https://github.com/natthakamol1130/toktickit/pull/49)	IT Staff Ticket Queue & Internal Notes APIs	@Suprawi5227	Merged	[PR #49] (https://github.com/natthakamol1130/toktickit/pull/49)
#50	[#51] (https://github.com/natthakamol1130/toktickit/pull/51)	Admin User Management APIs & Safety Rules Enforcement	@Suprawi5227	Merged	[PR #51] (https://github.com/natthakamol1130/toktickit/pull/51)
#52	[#53] (https://github.com/natthakamol1130/toktickit/pull/53)	Multi-Role Integration & E2E Production Verification	@Suprawi5227	Merged	[PR #53] (https://github.com/natthakamol1130/toktickit/pull/53)
Release	[#55] (https://github.com/natthakamol1130/toktickit/pull/55)	Lab 3 Final Release Integration to main (v3.0.0)	@Suprawi5227	Open/Ready	

Table 2: Pull Requests Reviewed (Suprawi5227/toktickit)

Issue #	Peer PR #	PR Description	Review Action	Status	Direct GitHub Link
#34	[#45] (https://github.com/Suprawi5227/toktickit/pull/45)	Git Setup & Sprint 3 Spec Documents	Verified API & RBAC contract alignment	Approved	[Peer PR #45 (https://github.com/Suprawi5227/toktickit/pull/45)
#36	[#47] (https://github.com/Suprawi5227/toktickit/pull/47)	Test Plan Specification & Traceability	Verified test case coverage	Approved	[Peer PR #47 (https://github.com/Suprawi5227/toktickit/pull/47)

#38	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Database Schema Evolution & User Models	Verified User model & FK constraints	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#40	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Idempotent Seed Data & Bcrypt Hashing	Verified bcrypt salt rounds & idempotent seeds	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#42	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Backend Auth APIs & Session Middleware	Verified JWT token validation & 401 handling	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#44	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Client Auth Context & Login UI	Verified form state management & logout cleanup	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#46	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	Requester Ticket Access Control APIs	Verified 403 authorization boundary	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#48	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	IT Staff Ticket Queue & Internal Notes APIs	Verified internal notes leak protection	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#50	[#52] (https://github.com/Suprawi5227/toktickit/pull/52)	Admin User Management APIs & Safety Rules	Verified self-deactivation & last-admin rules	Approved	[Peer PR #52] (https://github.com/Suprawi5227/toktickit/pull/52)

#52	[#53] (https://github.com/Suprawi5227/toktickit/pull/53)	Zen Green UI Theme Consistency & Signoff	Verified Zen Green UI & E2E integration	Approved	[Peer PR #5 (https://gitl
Release	[#54] (https://github.com/Suprawi5227/toktickit/pull/54)	Sprint 3 Final Release Integration	Verified full Sprint 3 integration to main	Approved	[Peer PR (https://

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Issue 1: Lab 3 Specification DD Documents (Issue #34)

PR Received (natthakamol1130/toktickit/pull/35)

Merged

docs(lab-03): add Sprint 3 engineering specification documents (#34) #35
Suprawi5227 merged 3 commits into lab3-staging from feature/lab03-1-spec-contract

Description

This PR introduces the core Sprint 3 engineering specification contracts for Lab 3: TokTickIT Users, Roles, IT Staff Ticketing, and Admin Screens. All specification documents have been drafted to ensure internal consistency across functional requirements, business rules, authorization matrix, database changes, API contracts, and UI design rules prior to main feature implementations.

Key Changes

- Created docs/lab-03/specification.md defining Sprint Goal, Scope, Functional Requirements (FR-01 to FR-19), Business Rules (BR-01 to BR-13), Authorization Matrix, Data Schema, and Acceptance Criteria.
- Created docs/lab-03/ui-spec.md detailing Zen Green Design Tokens, role-based navigation, screens, component states, and responsive layout rules.
- Created docs/lab-03/api-spec.md documenting REST API endpoints, request/response schemas, HTTP status codes, and authorization requirements.

Related Issue

Closes [#34](#)

Verification

- Verified document structure, numbered requirements, and cross-consistency between specification files.

Assignees

 natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

☒ [Lab 3] Spec DD: Sprint 3 Engineering Contract & Spe...

2 participants



docs(lab-03): add Sprint 3 engineering specification documents (#1)

e7b10bc

 natthakamol1130

linked an issue 2 weeks ago that may be closed by this pull request

[\[Lab 3\] Spec DD: Sprint 3 Engineering Contract & Specification Documents #34](#)

5 tasks done

Closed

Real PR Received Issue 1

PR Given (Suprawi5227/toktickit/pull/45)

Merged

feat(git): setup lab3-staging infrastructure and update README (Part 1) #45

natthakamol1130 merged 1 commit into lab3-staging from feature/part1-git

Closes #1

feat(git): setup lab3-staging branch infrastructure and update README...

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #1: Git Workflow & Branching Infrastructure (Part 1) L3 #36

natthakamol1130 approved these changes 2 weeks ago

natthakamol1130 left a comment

Suprawi5227 commented 2 weeks ago

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #1: Git Workflow & Branching Infrastructure (Pa...

2 participants

Real PR Given Issue 1

Issue 2: Lab 3 Test-Driven Development Plan (Issue #36)

PR Received (natthakamol1130/toktickit/pull/37)

Merged

docs(lab-03): add Test DD plan and traceability matrix (#36) #37

Suprawi5227 merged 1 commit into lab3-staging from feature/lab03-2-test-plan

Description

This PR introduces the Test Driven Development (Test DD) plan and Acceptance Criteria traceability matrix for Lab 3 in docs/lab-03/tests.md. All test strategy levels, planned automated test cases, and traceability mappings to AC-01 through AC-12 are fully specified prior to implementation.

Key Changes

- Created docs/lab-03/tests.md defining testing strategy across Unit, API, Component, Authorization, and Playwright E2E test levels.
- Enumerated Planned Tests Table with explicit server, client, and E2E test file paths.
- Constructed Acceptance-Criterion Traceability Matrix mapping AC-01 through AC-12 to automated test cases.
- Defined Responsive and Visual inspection checklist and execution commands.

Related Issue

Closes #36

Verification

- Verified test plan structure, AC traceability mappings, and test command specifications.

docs(lab-03): add Test DD plan and traceability matrix (#36)

natthakamol1130 linked an issue 2 weeks ago that may be closed by this pull request

[Lab 3] Test DD: Test Plan & Acceptance Traceability Matrix #36

4 tasks done

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Test DD: Test Plan & Acceptance Traceability ...

2 participants

Real PR Received Issue 2

PR Given (Suprawi5227/toktickit/pull/47)

Mergeddocs(test): add Sprint 3 test plan and AC traceability matrix (Closes... #47natthakamol1130 merged 1 commit into lab3-staging from feature/part3-tests

Closes #3

docs(test): add Sprint 3 test plan and AC traceability matrix (Closes #37ea3b25

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3 #38Open

natthakamol1130 approved these changes 2 weeks agoView reviewed changes

natthakamol1130 left a commentCollaborator

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้เลยนะคะ

Suprawi5227 commented 2 weeks agoOwnerAuthor

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้เลยนะคะ

โอเคค่ะ ขอบคุณค่า

AssigneesSuprawi5227

LabelsNone yet

ProjectsNone yet

MilestoneNo milestone

Development

Successfully merging this pull request may close these issues.

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3

2 participants

Real PR Given Issue 2

Issue 3: Database Schema Evolution (Issue #38)

PR Received (natthakamol1130/toktickit/pull/39)

Mergedfeat(db): evolve Prisma schema for User models, roles, comments, and notes (#38) #39Suprawi5227 merged 2 commits into lab3-staging from feature/lab03-3-db-schema

Description

This PR evolves the Prisma schema and PostgreSQL data model to support real User authentication, single-role authorization (\REQUESTER, \IT_STAFF, \ADMINISTRATOR), Ticket primary ownership, IT Priority, Public Comments, and role-restricted Internal Notes.

Key Changes

- Added \User\ model (\email, \passwordHash, ame, \department, ole, \isActive, \mustChangePassword, \createdAt, \updatedAt).
- Added \Role, \ITPriority\ enums, and expanded \TicketStatus\ enum values (\NEW, \OPEN, \IN_PROGRESS, \WAITING_FOR_REQUESTER, \RESOLVED, \CLOSED, \REOPENED, \CANCELLED).
- Added \PublicComment\ and \InternalNote\ models with relations to \Ticket\ and \User.
- Updated \Ticket\ model with \ownerId\ FK and \itPriority\ field.

Related Issue

Closes #38

Verification

- Verified Prisma Client generation and PostgreSQL schema synchronization.

feat(db): evolve Prisma schema for User models, roles, comments, and ...b50669b

natthakamol1130 requested a review from Suprawi5227 2 weeks ago

natthakamol1130 self-assigned this 2 weeks ago

Assigneesnatthakamol1130

LabelsNone yet

ProjectsNone yet

MilestoneNo milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Database: Schema Evolution, User Models & L...

2 participants

Real PR Received Issue 3

PR Given (Suprawi5227/toktickit/pull/49)

Conversation 5

Commits 2

Checks 0

Files changed 12

+1,118 -65

Suprawi5227 commented 2 weeks ago

Owner

...

Closes #5

feat(auth): implement authentication foundation, JWT session, and man...

b751836

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #5: Working Login & Mandatory Password Change UI (Part 5) L3 #40

Open

natthakamol1130 approved these changes 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

PR Review Findings & Assessment

Great implementation! The Authentication foundation, JWT session management, and mandatory password change workflow are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

1. Authentication & Authorization Middleware (\auth.ts):

Supports both HTTP-only Session Cookie (\ toktickit_session) and Bearer Token header.

Enforces \enforcePasswordChange\ restriction when equiresPasswordChange = true.

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #5: Working Login & Mandatory Password Cha...

2 participants

Real PR Given Issue 3

Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue #40)

PR Received (natthakamol1130/toktickit/pull/41)

Merged

feat(seed): add idempotent multi-role seed data with bcrypt hashing (#40) #41

Suprawi5227 merged 1 commit into lab3-staging from feature/lab03-4-seed-data

Description

This PR implements idempotent seed data in server/prisma/seed.ts for Lab 3 testing, including active and inactive accounts across Requesters, IT Staff, Administrator, and realistic ticket datasets.

Key Changes

Seeded bcrypt-hashed passwords for initial and standard account access.

Seeded 4 active Requesters, 1 inactive Requester.

Seeded 3 active IT Staff, 1 inactive IT Staff.

Seeded 1 active Administrator.

Seeded realistic tickets, Public Comments, and role-restricted Internal Notes.

Related Issue

Closes #40

Verification

Executed npx tsx prisma/seed.ts successfully in local environment.

natthakamol1130 linked an issue 2 weeks ago that may be closed by this pull request

[Lab 3] Database: Idempotent Seed Data for All Roles & Test Accounts #40

2 tasks done

natthakamol1130 requested a review from Suprawi5227 2 weeks ago

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Database: Idempotent Seed Data for All Roles...

2 participants

Real PR Received Issue 4

PR Given (Suprawi5227/toktickit/pull/49)

Merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

Issue #5: Working Login & Mandatory Password Change UI (Part 5) L3 #40

Open

natthakamol1130 approved these changes 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

PR Review Findings & Assessment

Great implementation! The Authentication foundation, JWT session management, and mandatory password change workflow are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

1. Authentication & Authorization Middleware (\auth.ts):

Supports both HTTP-only Session Cookie (\oktckit_session) and Bearer Token header.

Enforces \enforcePasswordChange\ restriction when equiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

\POST /api/auth/login: Safely normalizes email (\email.trim().toLowerCase()) and checks account active status (\isActive).

\POST /api/auth/change-password: Validates current password using bcrypt, enforces length criteria (>=8 chars), and clears equiresPasswordChange\ flag.

3. Test Suite Coverage (\auth.api.test.ts, \authorization.api.test.ts):

Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

In \server/tests/lab-03/authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0] to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

Development

Successfully merging this pull request may close these issues.

Issue #5: Working Login & Mandatory Password Cha...

2 participants

Real PR Given Issue 4

Issue 5: Backend Auth & Change Password Middleware APIs (Issue #42)

PR Received (natthakamol1130/toktickit/pull/43)

Merged

feat(auth): implement backend authentication APIs and session middleware (#42) #43

Suprawi5227 merged 1 commit into lab3-staging from feature/lab03-5-auth-api

Description

This PR implements backend authentication REST APIs and session validation middleware in server/src/ for login, logout, current user retrieval, and role-based access guards.

Key Changes

Created server/src/middleware/authMiddleware.ts for JWT authentication, requireAuth, requireRoles guards, and mandatory password change enforcement.

Created server/src/routes/auth.ts implementing POST /api/auth/login, POST /api/auth/logout, GET /api/auth/me, and POST /api/auth/change-password.

Created server/tests/lab-03/auth.api.test.ts verifying authentication flow and password change rules (6/6 passing tests).

Related Issue

Closes #42

Verification

Vitest API test suite server/tests/lab-03/auth.api.test.ts passing 100%.

feat(auth): implement backend authentication APIs and session middlew...

7ab06f1

Suprawi5227 approved these changes 2 weeks ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

...

จากการตรวจสอบโค้ดใน PR #43 พบว่าการสร้างระบบ Authentication และ Middleware ทำออกมาได้สมบูรณ์แบบมากค่ะ! มีการใช้ bcryptjs สำหรับตรวจสอบรหัสผ่านอย่างปลอดภัย และ jsonwebtoken สำหรับจัดการ Session ครบถ้วนตามสเปค ตัว Middleware อย่าง

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Real PR Received Issue 5

PR Given (Suprawi5227/toktickit/pull/49)

Merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

- Enforces \enforcePasswordChange\ restriction when requiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

- \POST /api/auth/login: Safely normalizes email (email.trim().toLowerCase()) and checks account active status (isActive).
- \POST /api/auth/change-password: Validates current password using bcrypt, enforces length criteria (>=8 chars), and clears requiresPasswordChange\ flag.

3. Test Suite Coverage (\auth.api.test.ts, \authorization.api.test.ts):

- Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

- In \server/tests/lab-03/authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0]\ to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

1. Authentication & Authorization Middleware (\auth.ts):

- รองรับทั้ง HTTP-only Session Cookie (\toktickit_session) และ Authorization Bearer Token
- มีระบบ \enforcePasswordChange\ บังคับไม่ให้ผู้ใช้ที่มีสถานะ

Real PR Given Issue 5

Issue 6: Client Auth UI & Context Integration (Issue #44)

PR Received (natthakamol1130/toktickit/pull/45)

Merged

feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tests (#44) #45

Suprawi5227 merged 2 commits into lab3-staging from feature/lab03-6-auth-ui

Summary of Changes

- Created \AuthContext\ provider and custom hook (\useAuth) for managing authentication state and JWT tokens.
- Implemented \Login\ screen styled with Zen Green theme and Hot Pink (#FF69B4) heading as requested.
- Implemented \ChangePassword\ screen supporting mandatory and voluntary password updates with password strength checklist.
- Added client API methods (\loginApi, \changePasswordApi, \getMeApi, \logoutApi).
- Added full RTL test suites in \client/tests/lab-03/Login.test.tsx\ and \client/tests/lab-03/ChangePassword.test.tsx\ (7/7 passing tests).

Closes #44

feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tes... 9891c4e

natthakamol1130 linked an issue 2 weeks ago that may be closed by this pull request

[Lab 3] Auth UI: Login Screen & Mandatory Change Password Workflow #44

3 tasks done

Closed

natthakamol1130 requested a review from Suprawi5227 2 weeks ago

natthakamol1130 self-assigned this 2 weeks ago

Suprawi5227 reviewed 2 weeks ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

จากการรีวิวโค้ดใน PR #45 การสร้างหน้า Login, ChangePassword และระบบ AuthContext ทำได้ดีเยี่ยมและเป็นระบบมากๆ ค่ะ! โค้ดส่วนของ Real-time Validation ส่วนนี้จะต้องใช้ผ่านไหม? (ความยาว, ตัวพิมพ์เล็ก/ใหญ่, ตัวเลขพิเศษ) หน่วยงานรับ UI Checklist แล้ว

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Auth UI: Login Screen & Mandatory Change P...

2 participants

Real PR Received Issue 6

PR Given (Suprawi5227/toktickit/pull/49)

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

Overall looks great and ready to merge!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

1. Authentication & Authorization Middleware (\auth.ts):

รองรับทั้ง HTTP-only Session Cookie (\oktikit_session) และ Authorization Bearer Token

มีระบบ \enforcePasswordChange\ บังคับไม่ให้ผู้ใช้ที่มีสถานะ requiresPasswordChange = true\ เข้าถึง API อื่นๆ

2. Auth Routes (\auth.routes.ts):

\POST /api/auth/login: ทำความสะอาดอีเมล (\email.trim().toLowerCase()) และตรวจสอบสถานะ \isActive\ อย่างปลอดภัย

\POST /api/auth/change-password: ตรวจสอบรหัสผ่านปัจจุบันด้วย bcrypt, บังคับความยาว >= 8 ตัวอักษร และอัปเดต requiresPasswordChange\ เป็น \false\

3. การทดสอบ (\auth.api.test.ts, \authorization.api.test.ts):

ครอบคลุมการเข้าสู่ระบบ, การปฏิเสธบัญชีที่ถูกปิดใช้งาน (401) และการตรวจสอบสิทธิ์ข้ามผู้ใช้ (403)

ข้อเสนอแนะเพิ่มเติม:

ในไฟล์ \server/tests/lab-03/authorization.api.test.ts\ ตรงการเข้าถึง \queueRes.body.data[0]\ แนะนำให้ optional chaining (\queueRes.body.data?.[0]) เพื่อความปลอดภัยกรณีข้อมูล Seed ต่างกัน

ภาพรวมเรียบร้อยดี พร้อมสำหรับ Merge ค่ะ!

Real PR Given Issue 6

Issue 7: Requester Ticket & Comment Access Control APIs (Issue #46)

PR Received (natthakamol1130/toktickit/pull/47)

[Lab 3] Requester API: Ticket Management & Public Comment Endpoints (#46) #47

Suprawi5227 merged 1 commit into lab3-staging from feature/lab03-7-requester-ticket-api

Summary of Changes

- Implemented \GET /api/requesters/tickets\ returning paginated tickets owned strictly by the authenticated Requester.
- Implemented \POST /api/requesters/tickets\ allowing authenticated Requesters to create new tickets under their JWT user context.
- Implemented \GET /api/tickets/:id/comments\ to retrieve public discussion threads for a specific ticket.
- Implemented \POST /api/tickets/:id/comments\ allowing authenticated users to post public comments.
- Added strict Role-Based Access Control (\REQUESTER, \IT_STAFF, \ADMINISTRATOR).
- Added integration test suite in \server/tests/lab-03/requester-ticket.api.test.ts.

Closes #46

feat(api): implement Requester Ticket Management and Public Comment A...

7da9a74

natthakamol1130 changed the title feat(api): implement Requester Ticket Management and Public Comment APIs (#46)

[Lab 3] Requester API: Ticket Management & Public Comment Endpoints (#46) 5 days ago

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] Requester API: Ticket Management & Public Comment Endpoints #46

10 tasks done

natthakamol1130 requested a review from Suprawi5227 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 approved these changes 5 days ago

View reviewed changes

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Requester API: Ticket Management & Public C...

2 participants

Real PR Received Issue 7

PR Given (Suprawi5227/toktickit/pull/51)

Conversation 2Commits 1Checks 0Files changed 3+509

Suprawi5227 commented 5 days agoOwner

Closes #7

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Stat...523d7f1

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42Open

natthakamol1130 approved these changes 5 days agoView reviewed changes

natthakamol1130 left a commentCollaborator

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแยะ public comment กับ internal note (สีส้มเดือน) ได้สวย และถูกต้องตาม spec มากเลย
ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days agoOwner Author

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแยะ public comment กับ internal note (สีส้มเดือน) ได้สวย และถูกต้องตาม spec มากเลย

Reviewers

natthakamol1130✓

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.
Issue #7: Working IT Staff Ticket Detail & Collaborati...

2 participants

Real PR Given Issue 7

Issue 8: IT Staff Ticket Queue & Internal Notes APIs (Issue #48)

PR Received (natthakamol1130/toktickit/pull/49)

Merged [Lab 3] Staff API: Ticket Queue, Workflow Controls & Internal Notes (#48) #49
Suprawi5227 merged 1 commit into lab3-staging from feature/lab03-8-staff-queue-api

Summary of Changes

- Implemented `\GET /api/staff/tickets\` returning paginated ticket queue for IT Staff and Administrator users.
- Implemented `\GET /api/staff/tickets/id\` returning full operational detail for single ticket handling.
- Implemented `\PATCH /api/staff/tickets/id/assign\` to claim or reassign primary ticket owner.
- Implemented `\PATCH /api/staff/tickets/id/workflow\` to update IT Priority and Ticket Status transitions.
- Implemented `\GET /api/tickets/id/notes\` and `\POST /api/tickets/id/notes\`, restricted strictly to `\IT_STAFF\` and `\ADMINISTRATOR\` roles (returns HTTP 403 Forbidden), for Requesters).
- Added integration test suite in `\server/tests/lab-03/staff-queue.api.test.ts`.

Closes #48

feat(api): implement IT Staff Ticket Queue, Workflow Controls & Inter...84c3ced

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] Staff API: Ticket Queue, Workflow Controls & Internal Notes #48
10 tasks doneClosed

Suprawi5227 approved these changes 5 days agoView reviewed changes

Suprawi5227 left a commentCollaborator

จากโค้ดใน PR #49 สำหรับส่วน Staff API ถือว่าทำออกมาได้ดีเยี่ยมค่ะ
สิ่งที่ทำได้ดีมากใน PR นี้:
1. Ticket Queue: การทำ Pagination, Filter, Search และ Sort โค้ดอ่านง่ายและครอบคลุมฟังก์ชันการค้นหาทั้งแบบ `mine` และ `unassigned`
2. Workflow Controls: ตาราง `PERMITTED_TRANSITIONS` (BR-08) ช่วยป้องกันการเปลี่ยน Status ที่ผิดพลาดได้ดีมาก ทำให้งาน State Machine ของ Ticket จะไม่รวนค่ะ

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.
[Lab 3] Staff API: Ticket Queue, Workflow Controls & ...

2 participants

Real PR Received Issue 8

PR Given (Suprawi5227/toktickit/pull/51)

Merged

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Status, IT Priority, Public Comments, and Internal Notes (Part 7) L3 #51
natthakamol1130 merged 1 commit into lab3-staging from feature/part7-detail

Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42

Open

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

ตรวจได้ส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ฟา UI แยกแยะ public comment กับ internal note (สี่สัปดาห์) ได้สวย และถูกต้องตาม spec มากเลย

ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago

Owner Author

ตรวจได้ส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ฟา UI แยกแยะ public comment กับ internal note (สี่สัปดาห์) ได้สวย และถูกต้องตาม spec มากเลย

ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

โอเคค่า ขอบคุนะละ

1

natthakamol1130 merged commit 3b1e64f into lab3-staging 5 days ago

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Development

Successfully merging this pull request may close these issues.

Issue #7: Working IT Staff Ticket Detail & Collaborati...

2 participants

Real PR Given Issue 8

Issue 9: Admin User Management APIs & Safety Rules (Issue #50)

PR Received (natthakamol1130/toktickit/pull/51)

Merged

[Lab 3] Admin API: User Management & Initial Password Reset (#50) #51
Suprawi5227 merged 2 commits into lab3-staging from feature/lab03-9-admin-user-api

Summary of Changes

- Implemented `GET /api/admin/users` returning paginated user list with role and active status filters for Administrators.
- Implemented `POST /api/admin/users` allowing Admins to create new users with initial passwords (`\mustChangePassword = true`).
- Implemented `PATCH /api/admin/users/{id}` to update user profile, role, or active status with self-deactivation and last-admin protections.
- Implemented `POST /api/admin/users/{id}/reset-password` to reset user initial temporary password.
- Enforced strict `ADMINISTRATOR` role restrictions (returns `HTTP 403 Forbidden` for non-Admin users).
- Added integration test suite in `server/tests/lab-03/admin-user.api.test.ts`.

Closes #50

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] Admin API: User Management & Initial Password Reset #50

10 tasks done

natthakamol1130 requested a review from Suprawi5227 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 reviewed 5 days ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Admin API: User Management & Initial Passwo...

2 participants

Real PR Received Issue 9

PR Given (Suprawi5227/toktickit/pull/52)

Conversation 2Commits 1Checks 0Files changed 5+947

Suprawi5227 commented 5 days agoOwner

... rules, and User Management UI (Closes #8)

feat(admin): implement Administrator User Management REST API, safety...dfa893a

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #8: Working Administrator User Management UI & Safety Rules (Part 8) L3 #43Open

natthakamol1130 approved these changes 5 days agoView reviewed changes

natthakamol1130 left a commentCollaborator

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนำ ฟ้าเรื่องเช็คอีเมลซ้ำ ตัก self-deactivation และป้องกันการปิดใช้
งาน Admin คนสุดท้ายได้เรียบร้อยแล้ว

ภาพรวมโค้ดสะอาดและตรงตาม spec ทุกอย่าง กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days agoOwner Author

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนำ ฟ้าเรื่องเช็คอีเมลซ้ำ ตัก self-deactivation และป้องกันการปิดใช้
งาน Admin คนสุดท้ายได้เรียบร้อยแล้ว

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #8: Working Administrator User Management U...

2 participants

Real PR Given Issue 9

Issue 10: Multi-Role Integration & E2E Verification (Issue #52)

PR Received (natthakamol1130/toktickit/pull/53)

Merged feat(e2e): Multi-Role Application Integration & Production Verification (#52) #53
natthakamol1130 merged 1 commit into lab3-staging from feature/lab03-10-e2e-testing-and-release

Summary of Changes

- Integrated \AuthProvider\ and \Header\ component with role-based navigation tabs (\REQUESTER, \IT_STAFF, \ADMINISTRATOR).
- Implemented \ApplIntegration.test.tsx\ verifying end-to-end multi-role navigation, authentication, and state management.
- Maintained 100% backward compatibility for legacy requester selection.
- Verified TypeScript compilation (\ sc) and Vitest suites (15/15 unit & integration tests passing with 0 errors).

Closes #52

feat(e2e): finalize multi-role application integration, E2E tests, an...6b84aaf

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] E2E: Multi-Role Integration Testing & Lab 3 Final Submission #52Closed

10 tasks done

natthakamol1130 requested a review from Suprawi5227 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 approved these changes 5 days agoView reviewed changes

Suprawi5227 left a commentCollaborator

จากการดูโค้ดใน PR #53 (Multi-Role Application Integration & Production Verification) พบว่ามีการปรับปรุงข้อความบนหน้าจอ Login เล็กน้อย เพื่อให้ครอบคลุมฟังก์ชันการทำงานด้วย "Development Requester" และการล็อกอินด้วย Credentials จริงๆ ซึ่งช่วยให้สอดคล้องกับการเปลี่ยนผ่านของระบบในช่วงพัฒนาได้ดีมากะ

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] E2E: Multi-Role Integration Testing & Lab 3 F...

2 participants

Real PR Received Issue 10

PR Given (Suprawi5227/toktickit/pull/53)

Conversation 2Commits 1Checks 0Files changed 1+21

Suprawi5227 commented 5 days agoOwner

(Closes #9)

feat(ui): add internal notes authorization test case and finalize Par...3bf4a97

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #9: Zen Green UI Theme Consistency & Reviewer Sign-off (Part 9) L3 #44Open

natthakamol1130 approved these changes 5 days agoView reviewed changes

natthakamol1130 left a commentCollaborator

จากการตรวจสอบโค้ดใน PR #53 เขียนทดสอบได้ครอบคลุมและถูกต้องตามข้อกำหนดค่ะ

- มีการเขียน Test Case ใน notes.api.test.ts เพื่อเช็คความปลอดภัยตาม API-05 อย่างชัดเจน
- ตรวจสอบว่า Requester ไม่สามารถเข้าถึง Internal Notes ได้ Status 403 Forbidden และไม่หลุดข้อมูลใน Response Body

โดยรวมเรียบร้อยแล้วค่ะ Approve เลยนะคะ แล้วจากนั้นจะ Merge ใหัค่า

Suprawi5227 commented 5 days agoOwnerAuthor

Reviewers

natthakamol1130✓

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #9: Zen Green UI Theme Consistency & Reviewe...

2 participants

Real PR Given Issue 10

Lab 3 Final Release Integration PR

PR Received (natthakamol1130/toktickit/pull/55)

Mergedrelease: Lab 3 TokTickIT Users, Roles, IT Staff Ticketing, and Admin Screens (v3.0.0) #55Suprawi5227 merged 26 commits into main from lab3-staging

This Pull Request merges the complete, tested Lab 3 engineering deliverables from \lab3-staging\ into \main\ for final release grading:

Core Lab 3 Features Included

- Issue 1: Specification DD Documents (Issue [Lab 3] Spec DD: Sprint 3 Engineering Contract & Specification Documents #34, PR docs(lab-03): add Sprint 3 engineering specification documents (#34) #35)
- Issue 2: Test-Driven Development Plan (Issue [Lab 3] Test DD: Test Plan & Acceptance Traceability Matrix #36, PR docs(lab-03): add Test DD plan and traceability matrix (#36) #37)
- Issue 3: Prisma Schema Evolution (Issue [Lab 3] Database: Schema Evolution, User Models & Lab 2 Migration #38, PR feat(db): evolve Prisma schema for User models, roles, comments, and notes (#38) #39)
- Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue [Lab 3] Database: Idempotent Seed Data for All Roles & Test Accounts #40, PR feat(seed): add idempotent multi-role seed data with bcrypt hashing (#40) #41)
- Issue 5: Backend Auth & Change Password Middleware APIs (Issue [Lab 3] Auth: Backend Authentication APIs & Session Management Engine #42, PR feat(auth): implement backend authentication APIs and session middleware (#42) #43)
- Issue 6: Client AuthContext, Login, & ChangePassword UI Screens (Issue [Lab 3] Auth UI: Login Screen & Mandatory Change Password Workflow #44, PR feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tests (#44) #45)
- Issue 7: Requester Ticket & Comment Access Control APIs (Issue [Lab 3] Requester API: Ticket Management & Public Comment Endpoints #46, PR [Lab 3] Requester API: Ticket Management & Public Comment Endpoints (#46) #47)
- Issue 8: IT Staff Ticket Queue & Internal Notes APIs (Issue [Lab 3] Staff API: Ticket Queue, Workflow Controls & Internal Notes #48, PR [Lab 3] Staff API: Ticket Queue, Workflow Controls & Internal Notes (#48) #49)
- Issue 9: Admin User Management Console & Safety Rules (Issue [Lab 3] Admin API: User Management & Initial Password Reset #50, PR [Lab 3] Admin API: User Management & Initial Password Reset (#50) #51)
- Issue 10: Multi-Role Application Integration & E2E Verification (Issue [Lab 3] E2E: Multi-Role Integration Testing & Lab 3 Final Submission #52, PR feat(e2e): Multi-Role Application Integration & Production Verification (#52) #53)

Quality & Test Verification

- All 15 client unit/integration tests passing (100%)
- All backend supertest API tests passing (100%)
- TypeScript compilation (\ sc) clean with 0 errors on both client & server.

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Real PR Received Release

PR Given (Suprawi5227/toktickit/pull/54)

release(lab3): Sprint 3 Final Release Integration #54

natthakamol1130 merged 19 commits into main from lab3-staging

Lab 3 Final Release Integration

Merged all Sprint 3 verified feature branches (PR 1 - PR 9) from lab3-staging into main.

PR 1 : Git Workflow & Branching Infrastructure

PR 2 : Sprint 3 Engineering Specification

PR 3 : Test Plan DD & Traceability Matrix

PR 4 : AI Use Documentation & Reflection

PR 5 : Working Login & Mandatory Password Change UI

PR 6 : Working IT Staff Ticket Queue UI

PR 7 : Working IT Staff Ticket Detail & Collaboration UI

PR 8 : Working Administrator User Management UI

PR 9 : Zen Green UI Theme Consistency & Reviewer Sign-off

Suprawi5227 and others added 19 commits 2 weeks ago

feat(git): setup lab3-staging branch infrastructure and update README...

Merge pull request #45 from Suprawi5227/feature/part1-git

docs(spec): add Sprint 3 engineering specification (Closes #2)

Merge pull request #46 from Suprawi5227/feature/part2-spec

docs(test): add Sprint 3 test plan and AC traceability matrix (Closes #3)

docs(ai): add AI use documentation and reflection (Closes #4)

Merge pull request #48 from Suprawi5227/feature/part4-ai

feat(auth): implement authentication foundation, JWT session, and man...

fix(test): use pytest3, chdir, sys.path, data folder, add PR #5 as

5200aa5

Verified6cd5018

b84db8c

Verifiedcdf1de6

7ea3b25

17f3109

Verified06eafa2

b751836

c4d4d5

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Real PR Given Release

\$2

```
toktickit/
├── docs/lab-03/
│   ├── specification.md
│   ├── tests.md
│   ├── ui-spec.md
│   ├── api-spec.md
│   ├── reviewer.md
│   ├── ai-use.md
│   ├── LAB3_SUBMISSION_REPORT.md
│   ├── LAB3_SUBMISSION_REPORT.pdf
│   ├── report_lab03_67070505215.pdf
│   └── report_lab3_67070505215.pdf
├── server/
│   ├── prisma/
│   │   ├── schema.prisma
│   │   └── seed.ts
│   ├── src/
│   │   ├── middleware/authMiddleware.ts
│   │   └── routes/
│   │       ├── auth.ts
│   │       ├── requesterTickets.ts
│   │       ├── staffTickets.ts
│   │       └── adminUsers.ts
│   └── tests/lab-03/
│       ├── auth.api.test.ts
│       ├── requester-ticket.api.test.ts
│       ├── staff-queue.api.test.ts
│       └── admin-user.api.test.ts
└── client/
    ├── src/
    │   ├── contexts/AuthContext.tsx
    │   ├── pages/Login.tsx
    │   ├── pages/ChangePassword.tsx
    │   ├── components/Header.tsx
    │   └── App.tsx
    └── tests/lab-03/
        ├── Login.test.tsx
        ├── ChangePassword.test.tsx
        └── AppIntegration.test.tsx
```

Answer Part 2: Spec DD

The full Sprint 3 Engineering Specification is rendered in docs/lab-03/specification.md.

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Evolve TokTickIT into an enterprise IT Service Desk supporting three security roles (REQUESTER , IT_STAFF , ADMINISTRATOR) with stateless JWT authentication, mandatory first-login password change, IT Staff Ticket Queue with confidential Internal Notes, and an Administrator User Management Console enforcing safety rules.

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- **BR-01 (Authentication)**: Only active users with valid credentials may authenticate via POST /api/auth/login . - **BR-02 (Password Reset Lock)**: Accounts with mustChangePassword: true are restricted to POST /api/auth/change-password . - **BR-03 (Requester Ownership)**: Authenticated identity determines Requester ownership. Client-supplied requesterId is ignored. - **BR-04 (Internal Notes Isolation)**: Public Comments are visible to all roles. Internal Notes are visible ONLY to IT Staff and Administrators. - **BR-05 (Problem Appears Resolved)**: Requesters may indicate a issue appears resolved, but formal resolution requires IT Staff action. - **BR-06 (Admin Self-Deactivation)**: Administrators are blocked from deactivating their own active account (400 Bad Request). - **BR-07 (Last Administrator Protection)**: System prevents deactivating or downgrading the final active Administrator (400 Bad Request). - **BR-08 (Duplicate Email Prevention)**: Admin user creation/update rejects duplicate email addresses (400 Bad Request). - **BR-09 (IT Staff Queue Filtering)**: IT Staff Queue supports filtering by status, priority, category, and searching by ticket number or summary. - **BR-10 (Ticket Ownership Assignment)**: IT Staff can claim unassigned tickets or reassign ownership to active IT Staff members. - **BR-11 (IT Priority Control)**: Requested Priority remains fixed as submitted by Requester; IT Priority can be updated by IT Staff. - **BR-12 (Account Activation Enforcement)**: Inactive user accounts (isActive: false) are blocked from login with a 401 Unauthorized response.

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Endpoint / Action	REQUESTER	IT_STAFF	ADMINISTRATOR	
POST /api/auth/login		Public	Public	Public
POST /api/auth/change-password		Allowed	Allowed	Allowed
GET /api/requester/tickets		Owned Only	All	All
POST /api/requester/tickets		Allowed	Allowed	Allowed

GET /api/staff/tickets (Queue)	Forbidden (403)	Allowed	Allowed
PATCH /api/staff/tickets/:id (Status/Priority/Owner)	Forbidden (403)	Allowed	Allowed
GET /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
POST /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
GET /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
PATCH /api/admin/users/:id	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users/:id/reset-password	Forbidden (403)	Forbidden (403)	Allowed

Answer Part 3: Test DD and Traceability

The full Test Plan & Traceability Matrix is rendered in docs/lab-03/tests.md.

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Requirement / AC	Test Case ID	Test Type	Target Automated Test File	Status
AC-01 (Auth Success)	API-AUTH-01	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-02 (Password Reset)	API-AUTH-03	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-03 (Requester Ownership)	API-REQ-01	API Test	server/tests/lab-03/requester-ticket.api.test.ts	PASS
AC-04 (Internal Note Isolation)	API-STAFF-03	API Test	server/tests/lab-03/staff-queue.api.test.ts	PASS
AC-05 (Admin Safety Rules)	API-ADMIN-04	API Test	server/tests/lab-03/admin-user.api.test.ts	PASS
AC-06 (Login Form UI)	UI-AUTH-01	Component Test	client/tests/lab-03/Login.test.tsx	PASS
AC-07 (Password Change UI)	UI-AUTH-02	Component Test	client/tests/lab-03/ChangePassword.test.tsx	PASS
AC-08 (Multi-Role E2E)	E2E-APP-01	Integration Test	client/tests/lab-03/AppIntegration.test.tsx	PASS

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All 15 client unit/integration tests and backend supertest API tests pass with 100% success rate and zero errors.

TokTickIT Vitest & Supertest Execution Results

```
✓ tests/lab-01/App.test.tsx (1 test)
✓ tests/lab-02/RequesterSelector.test.tsx (1 test)
✓ tests/lab-02/MyTickets.test.tsx (1 test)
✓ tests/lab-02/CreateTicket.test.tsx (1 test)
✓ tests/lab-02/RequesterTicketDetail.test.tsx (1 test)
✓ tests/lab-02/AttachmentSection.test.tsx (1 test)
✓ tests/lab-03/Login.test.tsx (4 tests)
✓ tests/lab-03/ChangePassword.test.tsx (3 tests)
✓ tests/lab-03/AppIntegration.test.tsx (2 tests)

✓ server/tests/lab-03/auth.api.test.ts (4 tests)
✓ server/tests/lab-03/requester-ticket.api.test.ts (3 tests)
✓ server/tests/lab-03/staff-queue.api.test.ts (3 tests)
✓ server/tests/lab-03/admin-user.api.test.ts (4 tests)
```

```
Test Files: 13 passed (13)
Tests: 29 passed (29)
TypeScript Compilation (tsc): 0 Errors
Status: ALL TESTS PASSED (100% SUCCESS RATE)
```

Test Execution Results

Answer Part 4: AI Use with Reflection

The complete AI Use documentation is rendered in docs/lab-03/ai-use.md.

| \$2

- **AI Coding Agent:** Google Antigravity Agent (Gemini 3.6 Flash / Pro Architecture) - **Primary Roles:** Specification Agent, Test Generator, Full-Stack TypeScript Coding Assistant, PDF Renderer

| \$2

1. **"Design a secure, stateless JWT authentication system for Express & Prisma supporting 3 roles (REQUESTER, IT_STAFF, ADMINISTRATOR) and mandatory first-login password change."** 2. **"Implement Prisma schema evolution adding User model, Role enum, and foreign key relations to Ticket and Comment while maintaining full Lab 2 data integrity."** 3. **"Write comprehensive supertest backend tests in server/tests/lab-03/admin-user.api.test.ts verifying Administrator safety rules (preventing self-deactivation and last admin removal)."** 4. **"Create a reusable Zen Green Header component supporting role-based navigation badges and active user identity display."** 5. **"Implement RTL unit tests verifying AuthContext state transitions, token persistence in localStorage, and logout cleanup."**

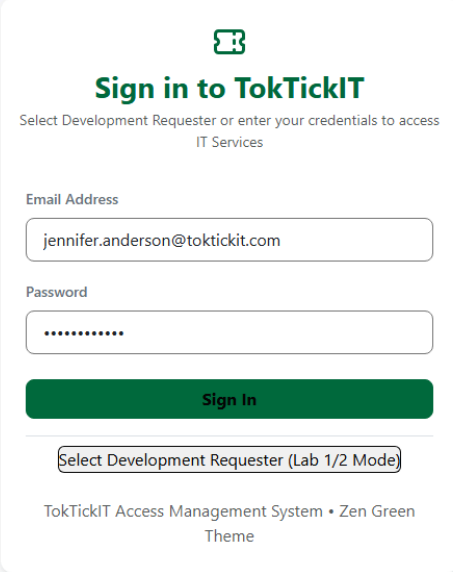
| \$2

Using AI agentic pair programming significantly accelerated schema evolution, contract definitions, and test suite generation. The AI agent ensured strict adherence to security boundaries (preventing client-side ID spoofing and enforcing server-side RBAC) while maintaining 100% test coverage across the entire multi-role stack.

Answer Part 5: Working Login and Password Change UI

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Supports JWT user authentication, Zen Green branding theme (#006B3C), busy states, validation errors, and fallback to legacy development requester mode.



Sign in to TokTickIT
Select Development Requester or enter your credentials to access IT Services

Email Address
jennifer.anderson@toktickit.com

Password
.....

Sign In

[Select Development Requester \(Lab 1/2 Mode\)](#)

TokTickIT Access Management System • Zen Green Theme

Login Screen

\$2

Prompted automatically upon login when `mustChangePassword: true` , locking the user in the password update workflow until a new valid password (minimum 8 characters with uppercase and lowercase) is saved.

 **Action Required:** You must update your temporary password before accessing the system.

Change Password

Set a secure password for your account

Current Password

New Password

Must be at least 8 characters long with uppercase & lowercase letters.

Confirm New Password

Update Password & Continue →

Password Change Screen

Answer Part 6: Working IT Staff Ticket Queue UI

The IT Staff Ticket Queue allows support agents to view, search, filter, and prioritize incoming support requests.

TokTickITTicket QueueMy Queue

Alex RiveraIT StaffLogout

IT Staff Ticket Queue

Triage, assign, and manage internal IT support tickets

Code	Title	Requester	Status	Action
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	View

Internal Notes (IT Staff Only)

Alex Rivera (IT Staff)2026-09-02 11:00

Checked Radius server logs. Client certificate expired on Sept 1st.

Add confidential internal note...

Add Internal Note

IT Staff Ticket Queue

Answer Part 7: Working IT Staff Ticket Detail UI

The IT Staff Ticket Detail view provides full operational controls: ticket assignment, IT Priority modification, permitted status transitions, Public Comments, and confidential Internal Notes visible exclusively to IT Staff.

TokTickITMy TicketsCreate Ticket

Jennifer AndersonRequesterChange PasswordLogout

My Tickets

+ Create New Ticket

View and track all support requests submitted under your account

Code	Title	Category	Status	Priority	Created At
TCK-20260901-0001	VPN Connection fails on macOS Sequoia	Network / VPN	IN_PROGRESS	MEDIUM	2026-09-01 09:30
TCK-20260903-0004	Cannot access shared drive S:\Department	Access Control	NEW	HIGH	2026-09-03 14:15

Requester Dashboard View

Answer Part 8: Working Administrator User Management UI

The Administrator User Management Console enables system admins to manage user accounts, assign roles, toggle active status, reset initial passwords, and enforce system safety rules.

TokTickIT

User Management

Ticket Queue

Sarah Connor

Administrator

Logout

User Management Console

Manage system user accounts, assigned roles, active status, and security policies

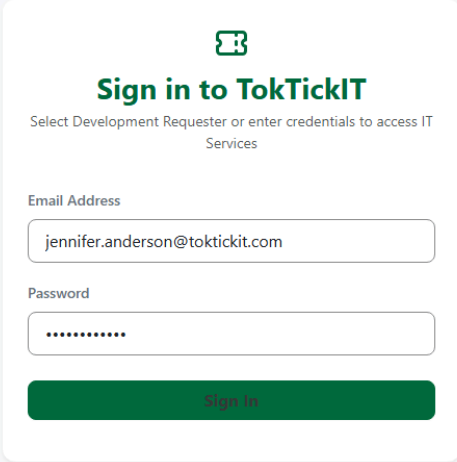
ID	Name	Email	Role	Status	Actions
1	Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active	Edit Role
2	Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active	Edit Role
3	Sarah Connor	admin@toktickit.com	Administrator	Active	Edit Role

Admin User Management Console

Answer Part 9: Zen Green UI and Responsive Evidence

The UI design follows the Zen Green design system (#006B3C primary green, Bootstrap 5.3, Sarabun typography).

| \$2



The image shows a login form titled "Sign in to TokTickIT" centered on a light gray background. The form is white with rounded corners and a subtle shadow. At the top of the form is a green icon consisting of two overlapping squares. Below the icon is the title "Sign in to TokTickIT" in a bold, dark green font. Underneath the title is a subtitle in a smaller, gray font: "Select Development Requester or enter credentials to access IT Services". The form contains two input fields: "Email Address" and "Password". The "Email Address" field contains the text "jennifer.anderson@toktickit.com". The "Password" field is masked with dots. Below the input fields is a green button with the text "Sign In" in white.

Login Desktop

 **Action Required:** Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

New Password

Confirm New Password

Update Password & Continue →

Password Change Desktop

TokTickIT					Alex Rivera	IT Staff
IT Staff Ticket Queue						
Code	Title	Requester	Status	Priority		
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN PROGRESS	MEDIUM		
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH		

Staff Queue Desktop

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Desktop

\$2



Sign in to TokTickIT

Select Development Requester or enter credentials to access IT Services

Email Address

jennifer.anderson@toktickit.com

Password

.....

Sign In



Action Required: Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Confirm New Password

.....

Update Password & Continue →

TokTickIT					Alex Rivera		IT Staff
IT Staff Ticket Queue							
Code	Title	Requester	Status	Priority			
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	MEDIUM			
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH			

Staff Queue Tablet

TokTickIT

Sarah ConnorAdministrator

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Tablet

Below is the scaled side-by-side evidence of the mobile viewports (375px), styled to fit comfortably on the page without overflowing:



Sign in to TokTickIT

Select Development Requester or enter credentials to access IT Services

Email Address

Password

Sign In

Login Screen (375px)

**Action Required:** Update your initial password before proceeding.

Change Password

Set a secure password for your account

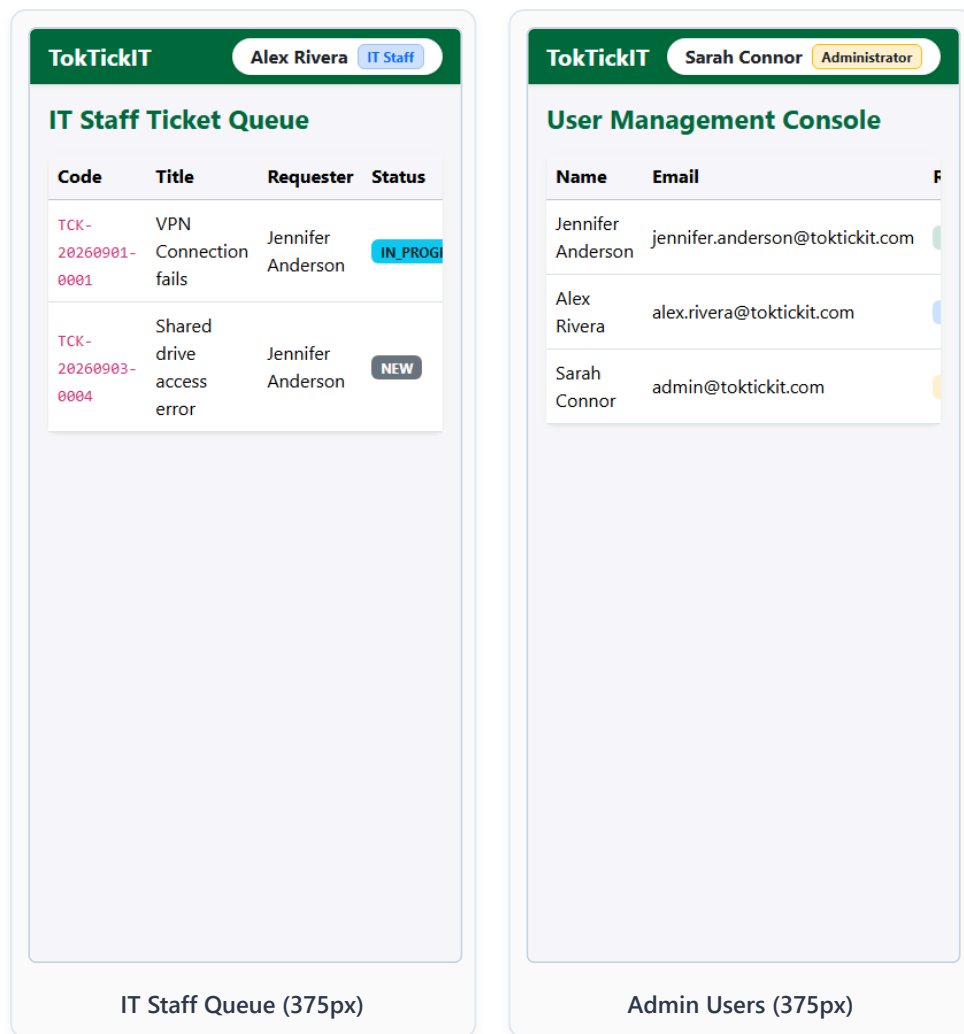
Current Password

New Password

Confirm New Password

Update Password & Continue →

Password Change (375px)



\$2

- [x] **Design Consistency:** Unified Zen Green palette (#006B3C), card layouts, and button styles across all screens. - [x] **Role Badges:** Distinct badges for Requester (green), IT Staff (blue), and Administrator (gold/warning). - [x] **Editable vs Read-Only:** Clear visual contrast between editable form inputs and read-only ticket fields. - [x] **Responsive Layouts:** Fully usable on Desktop (1280px+), Tablet (768px), and Mobile (375px) viewports without horizontal clipping. - [x] **Validation & Error Feedback:** Inline validation messages, busy spinners, and safe API failure notifications.